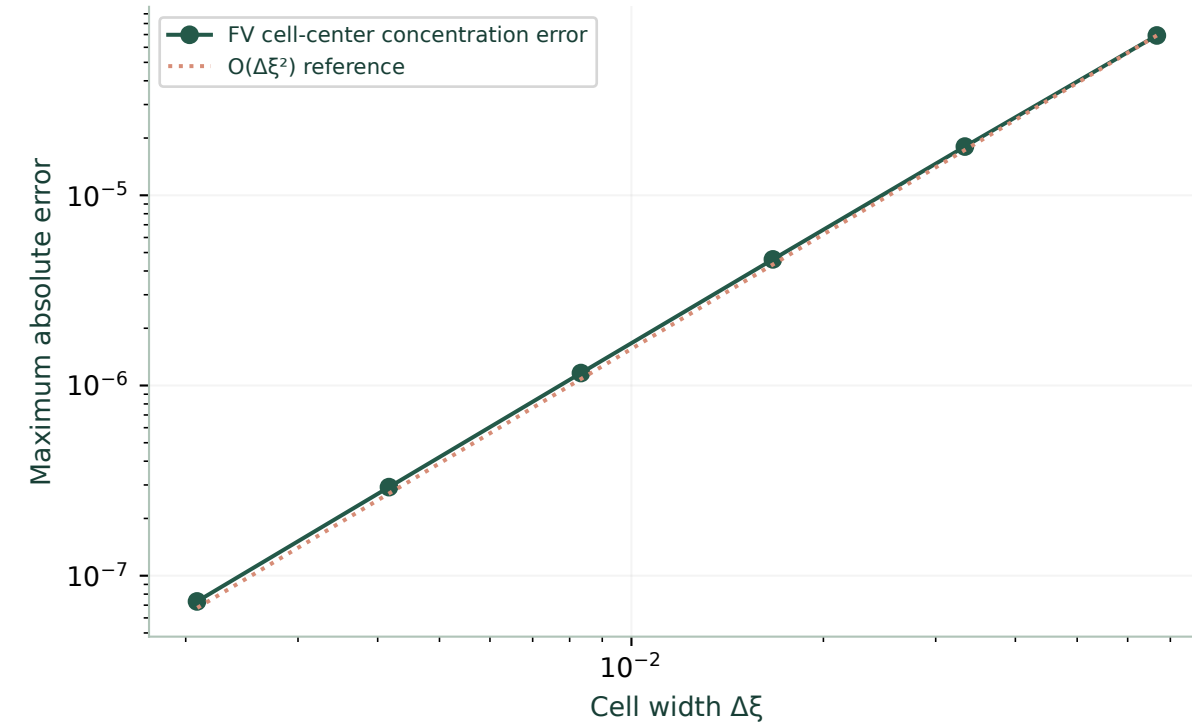
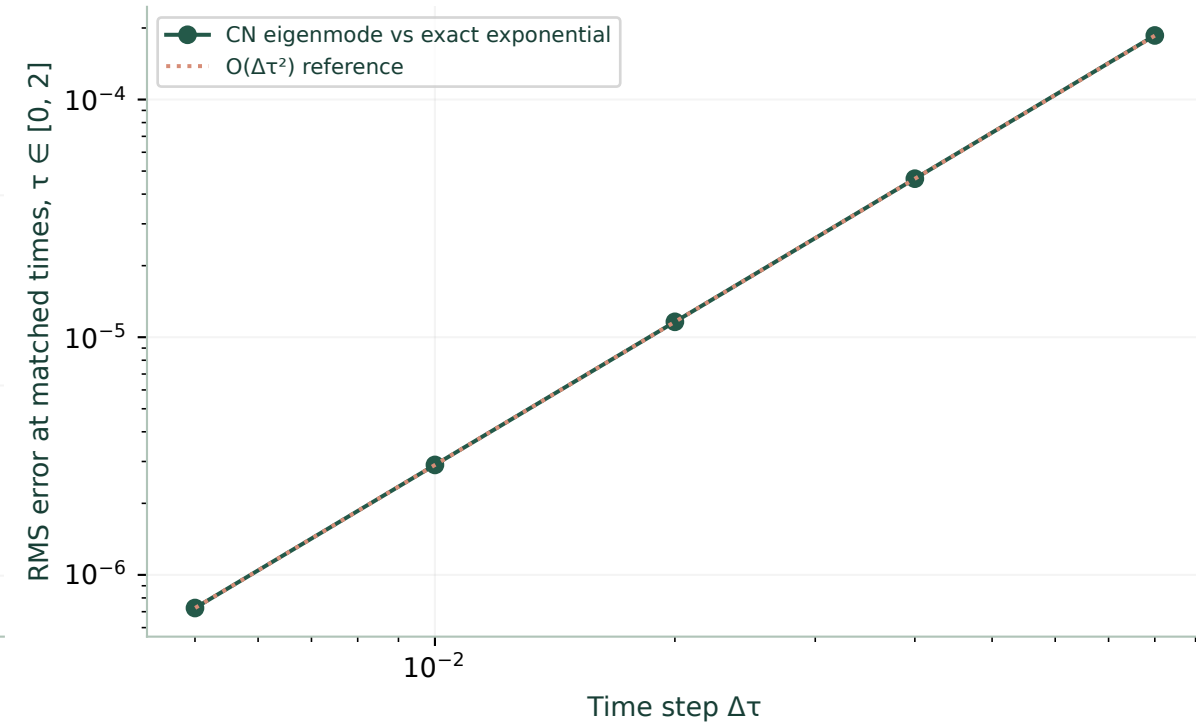


Convergence and independent limiting-case checks

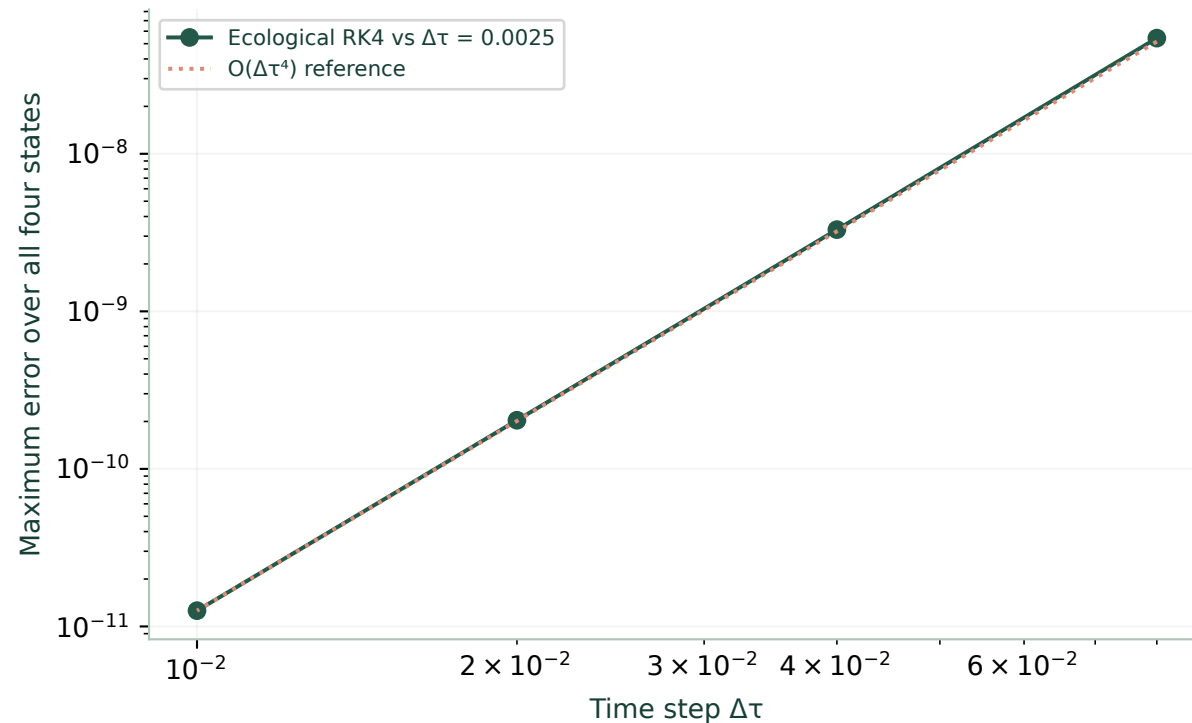
A Second-order spatial convergence



B Second-order time convergence for a smooth mode



C Pulse-aligned fourth-order integration



D Verification checks (not biological validation)

100 equilibrium comparisons; away-boundary max error

$1.87\text{e-}10$

Max step-wise FV conservation residual $2.24\text{e-}16$

Cumulative balance residual $1.00\text{e-}13$

Linear source scaling residual $0.00\text{e+}00$

Zero source $\rightarrow$ zero product; $B_i = 0 \rightarrow$ zero outflow

No EcN $\rightarrow$ no population-dependent Elafin source

$\gamma = 0 \rightarrow$ support/control populations identical

$Da \rightarrow 0$ agrees with the linear steady profile